CS341 Task Tracker – Test Report

CS341 Project | Evidence-aware verification | 1 November 2026

1. Test Strategy

Tests are divided into three evidence classes: (1) executable offline checks that do not require JDBC libraries, (2) JUnit 5 integration tests using the Java SQLite JDBC driver, and (3) manual Swing desktop acceptance tests. Results are not aggregated into a single pass count because these classes test different components.

2. Automated Test Inventory

Group	Cases	Execution evidence
Java	Maven compiles application code with `--release 11`.	PASS
JUnit/JDBC	Registration/login; task creation and assignment; creator/assignee restrictions; completion toggling; persistence; SQL literal safety; date/title bounds; direct guarded SQL mutation; reassignment and invalid input.	17 PASS; 0 fail/error/skip
Packaged JAR	Load SQLite JDBC from the shaded JAR and create/insert/read a temporary database.	PASS
Python SQL	Schema init/idempotency; foreign keys and index; unique usernames; role selection; priority; guarded updates; reopen persistence.	10/10 PASS; independent SQLite
PlantUML	Render seven canonical diagrams to SVG and PNG.	7/7 PASS
GUI	Registration, login, task CRUD, filters, sorting, role denial, restart persistence.	Manual run pending

3. Acceptance Scenario Matrix

ID	Procedure / Expected result	Observed status
AT-01	Register Alice; log out; log in with correct credentials.	Manual test pending
AT-02	Alice assigns a task to Bob. Task appears for creator and assignee.	Manual test pending
AT-03	Carol cannot view Alice-to-Bob task.	Manual test pending
AT-04	Bob completes and reopens the task; changes persist.	Manual test pending
AT-05	Bob cannot edit or delete Alice's task.	Manual test pending
AT-06	Alice edits, reassigns, and deletes her task.	Manual test pending
AT-07	Restart with the same `tasks.db`; data remains.	Manual test pending
AT-08	Use priority, due date, filter and sorting features.	Manual test pending

4. Executed and Missing Evidence

The successful command was `mvn -B clean verify`. The environment was Windows 11 with JDK 22.0.1 and Maven 3.9.9; the Maven compiler targets Java 11. The run executed all 17 JUnit methods: 0 failures, 0 errors and 0 skipped. The console log is `docs/evidence/maven-clean-verify.log`; Surefire XML and TXT reports are in `target/surefire-reports/`.

The first clean verification passed all tests but failed once during JAR assembly. The diagnostic package retry and the subsequent clean verification succeeded; the initial log is retained and not presented as a passing run. A separate smoke program confirmed that the packaged JAR can load SQLite JDBC and perform a database round trip. The independent Python check passed 10/10, but it does not replace JDBC/JUnit evidence.

All seven PlantUML sources were rendered using PlantUML 1.2025.10. XeLaTeX build logs and PDF text/page preflight are stored under `docs/evidence/latex/`. A real interactive Swing GUI run, screenshots, GitHub Actions run and team presentation have **not** been performed.

5. Completion Gate

Before submission, run AT-01–AT-08 in a graphical desktop session, record genuine GUI evidence, and provide authentic GitHub CI and student contribution evidence. The verified release ZIP does not satisfy these remaining human/remote requirements.